

Date: 21/7/2022

Thursday

General Science

Topic :-

Matter & its
States

∴ Matter :-

Any thing that
has weight and occupies
space is called matter.

There are some states
of matter.

- | | |
|-------------|--------------|
| (i) Solid | (iii) Gas |
| (ii) Liquid | (iv) Plasma. |

(i) Solid

Any thing which have fixed shaped, size, weight and volume is called solid.

Rock/Wood

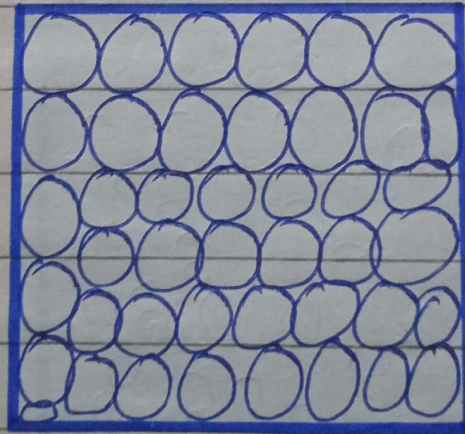
Types of Solid:-

- (i) Crystalline Solid
- (ii) Amorphous Solid
- (iii) Polymeric Solid
- (iv) Brittle Solid
- (v) Ductile Solid.

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(i) Crystalline Solid:-

Arrangement of atoms in these solids is very close and Regular. Same unit cells is repeated again and again in 3D Pattern. For example:- All metals.

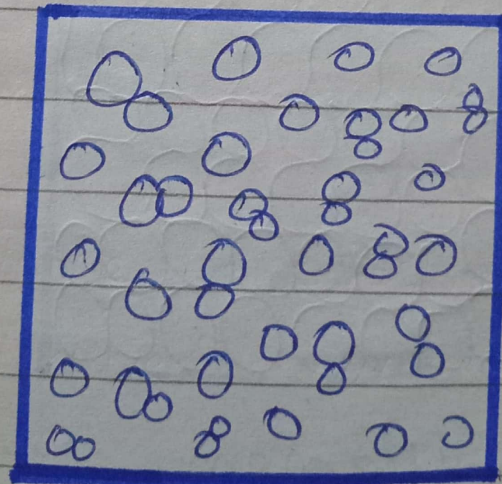


Diagram

(ii) Amorphous Solids:

Amorphous Solids means disorder. All those solids which have no ordered structure are called amorphous solid.

For example: Glass



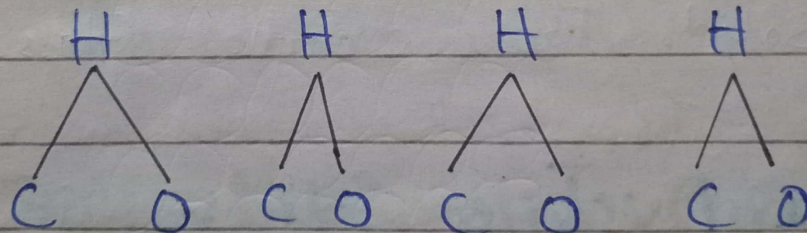
Diagram

(iii) Polymeric Solids:-

The ~~or~~ structure of these solids is between order & disorder.

For example:- Polystyrene, Polythene.

-: Structure:-



∴ On the basis of strength:
Brittle and Ductile
these are the two
types on the basis
of strength.

(1) Brittle solid:-

Those substance
which have little
strength is called Brittle
Solid. For example: Glass.

(2) Ductile Solid:-

Those substance
which have hard

Strength is called
Ductile solid.

e.g.: Gold / Silver

(2) Liquid:-

Any thing which
have fixed weight, volume
but no fixed shape
is called Liquid. In
Liquids, atoms are at
distance from each
other. We can easily
pour liquid in every
thing.

(iii) Gases

Any thing which have fixed weight but not fixed shaped and volume is called Gas. Gases has random motion.

(iv) Plasma:-

The state of matter in which positive and negative ions are present at very high Temperature is called plasma.

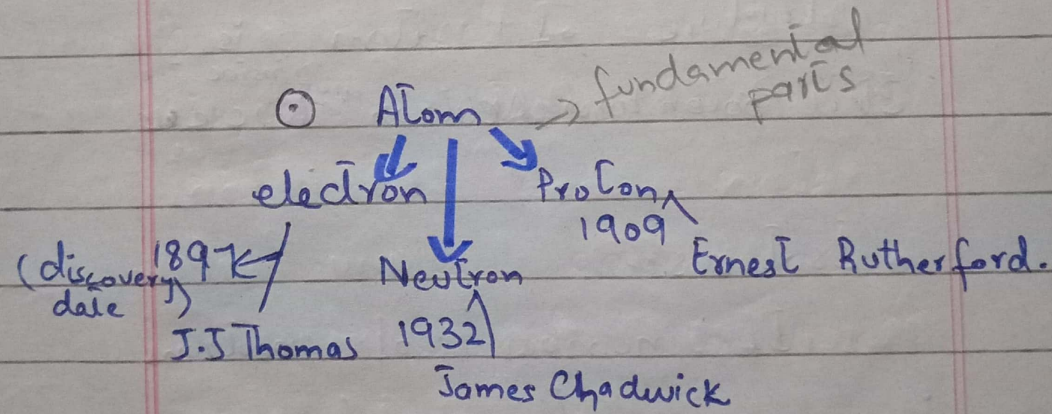
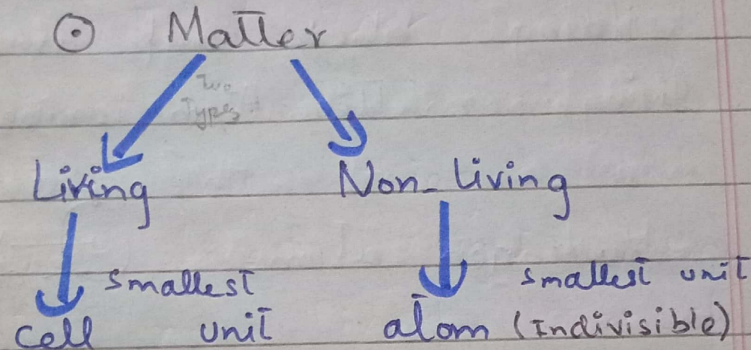
light

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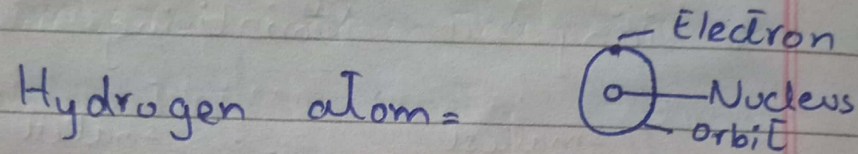
Atom:-

Matter



Atom:-

It is the smallest building block of non-living matter. Atom is neutral.



Structure of atom is 3D. Proton and Neutron ^{are} inside the Nucleus. Electron revolves around the Nucleus in shell, sub shell or orbitals.

Isotopes:-

If number of neutrons in any atom of element is greater

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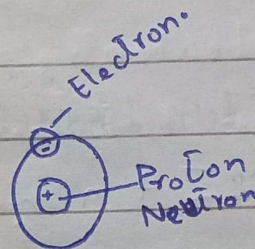
smallest particle of atom

Quarks

The weight of proton is 1837 larger than Electron.

than the number of proton in that element than such an atom is called isotopes of that atom.

In isotopes, atomic mass increase.



Total 118 elements
16 bioelements form body of human
92 Natural elements

26

mass number | H element.
atomic |
no.

∴

electron

atom

atom

atom

atom

atom

atom

atom

atom

no. of A
Proton + Neutron X element.
no. of Z proton.

∴

$$A = Z + n$$

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k and so on.

Atomic Number:- (Z)

Number of
protons in any atom
defines the atomic
number of that atom.

Mass Number:- (A)

$\therefore A = Z + n$ - number of neutron
Total Number of
proton and Neutrons
in Nucleus of any atom
describes the mass
number of that atom.

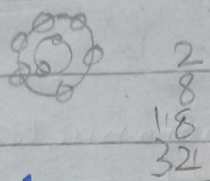
No. of Shell = $k = 1$

Formula.

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No of electrons = $2n^2$

S/Q



How many
electron

Molecules:-

can be

found If Two or more
in 4th than Two (atoms elements)
and 5th shell. of same elements

= $2n^2$ or different elements
= $2(4)^2$ are combined
= 32 electrons together chemically,

5th shell then they form a

= $2n^2$ For example :-

= $2(5)^2$ O_2 , H_2 , N_2 , H_2O , CO_2 ,

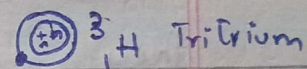
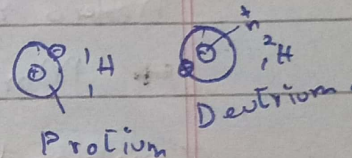
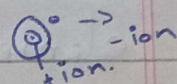
= 2×25 CO etc.

= 50.

→ If atoms are of
same elements then they

Information

Temperature of
Sun on the = 6000



Isotopes of
Hydrogen.
molecule.

Element is a substance made up of same type of atoms, having same atomic number.

H₂ H₂

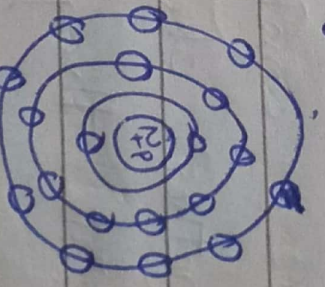
for homonuclear molecules.

→ If ^{H₂O} atoms are of different element then they form hetero nuclear molecules.

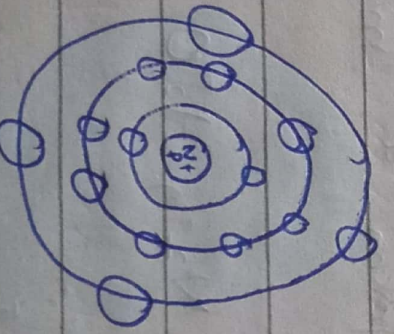
→ More than two atoms

combined chemically than they form polynuclear molecules.

C₆H₁₂O₆



Ar
gon
18



$2(2)^2$
 $\frac{2 \times 4}{8}$

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NaOH

(Na_2CO_3)
 NaHCO_3

Saturday

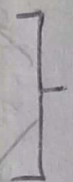
General Science.

Matter

phys. | chemically

Mixture

Element compound



Homogeneous Heterogeneous

Mixture Mixture.

Mixtures.

When

Two or more substances (pure form of matter)

than two elements are mixed

together physically and the

constituents maintain/retain

their identity then it is

called Mixture.

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When substances are mixed physically, mixtures are formed.

Types of Mixture:

- (i) Homogenous Mixture.
- (ii) Heterogenous Mixture.

(i) Homogenous Mixture.

When substance lose their identity in a mixture, then such a mixture is called homogenous mixture.

i.e - Sugar syrup, Salt syrup.

(ii) Heterogenous Mixture.

If constituents of a mixture don't lose their identity after mixing

up physically, then such a mixture is called heterogeneous mixture.

For example:- Fruit Salad,

Sand in water etc.

Both Homogenous & Heterogeneous processes for separation.

- (i) filtration
- (2) crystallization
- (3) Evaporation.

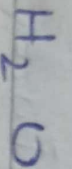
Compounds:-

Two

When they form more than two elements are combined chemically then they form compounds.

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For example: Hydrogen and oxygen make water. H_2O



2:16 fixed ratio by mass.
1:8

$\therefore$ The elements of compounds can't be separated.

Difference Mixture

Mixture

When substance are combined / mixed physically they form mixtures.

Compound

When elements are combined chemically in a fixed ratio by mass compounds are formed.

(2) Substances in

mixtures can be separated by filtration, evaporation,

(3)

Substances ~~can~~ ^{are not} mixed in a fixed ratio by mass in mixtures.

(4)

Mixtures can be homogeneous & heterogeneous.

(5)

Constituents of mixtures maintain their identity.

(2) Elements in

compounds can ^{not} be separated.

(3)

Elements are mixed in a fixed ratio by mass in compounds.

(4) Compounds can

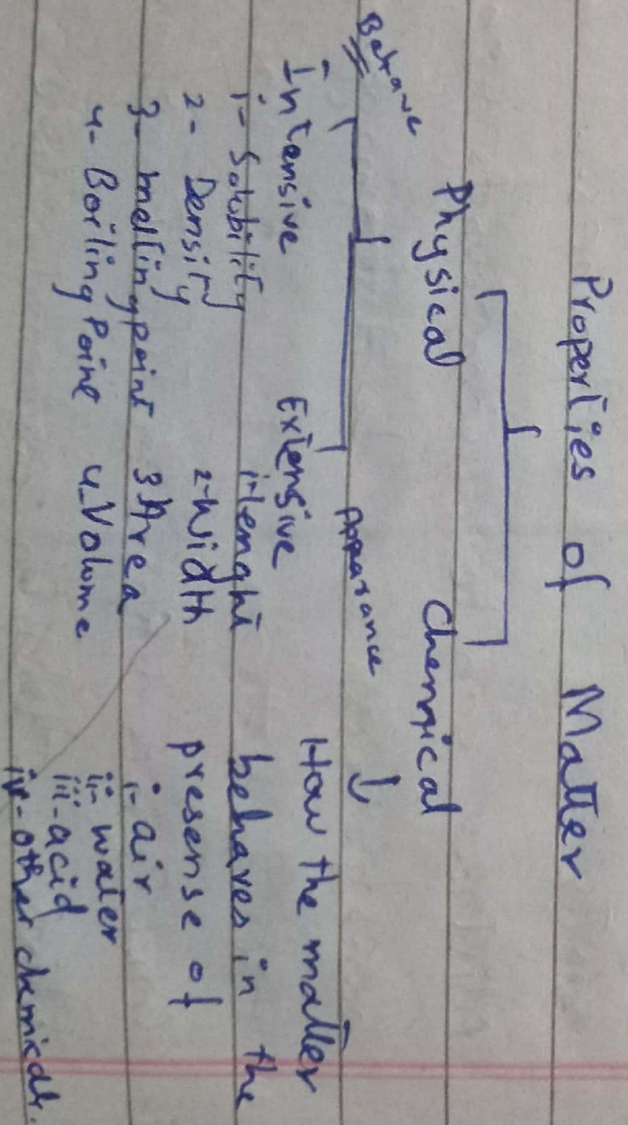
be homogeneous.

(5)

Constituents of compounds lose their identity.

(6) Mixture do not have any formula. sharp / fixed melting point.	(5) Compound. are represented by chemical formula. sharp melting points.
(1) Mixture don't have sharp / fixed melting point.	(2) Compound have sharp melting points.

Physical Properties of Matter:



Physical Properties:

Intensive Properties:

(i) Absorption of electromagnetic

The way photon energy

is taken up by matter.

(ii) Albedo

Reflecting power of

a surface.

(iii) Angular momentum:

The amount of rotation of an object.

(iv) Brilliance:

Tendency of a material break under stress.

Electromagnetism
from sun in the
rays
(i) Radio waves
(ii) Ultra violet
radiations
(iii) Visible light
(iv) Infrared
rays
(v) X ray
(vi) & rays.

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Boiling 100°C
Melting more than 0°C
Freezing 0°

(v) Boiling point:-

The Temperature where a liquid turns into gas / vapour

(vi) Melting point:-

The Temperature where a solid is converted to liquid.

(vii) Freezing point:-

The Temperature where a liquid converted to solid.

(viii) Capacitance:-

Capacity of a material to store charge.

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(ix) Concentration

Amount of any/one substance into a mixture.

(x) Density:

Mass per unit volume is density.

Formula: $D = \frac{M}{V}$

(xi) Dielectric constant:-

Storage of electric and magnetic energy is called Dielectric constant.

(xii) Ductility:-
Ability of a substance to be stretched into wires is called ductility.

Clues of Chemical Reaction:-

- (i) Irreversible Reaction
- (ii) Colour change
- (iii) Bubbles formation
- (iv) Gases are produced.

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Chemical Reaction:

Elements Name:

Oxygen $\rightarrow$ O

Chlorine $\rightarrow$ Cl

Magnesium $\rightarrow$ Mg

Nitrogen $\rightarrow$ N

Hydrogen $\rightarrow$ H

Boron $\rightarrow$ B

Germanium $\rightarrow$ Ge

Silicon $\rightarrow$ Si

Bromine $\rightarrow$ Br

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Mercury $\rightarrow$ Hg
Gold $\rightarrow$ Au
Iron $\rightarrow$ Fe

} Due to Latin language

-: Molecular formula:

3 Elements
have

Copper Sulphate $\rightarrow$ $CuSO_4$

Water $\rightarrow$ H_2O

Salt $\rightarrow$ NaCl

Hydrochloric Acid $\rightarrow$ HCl

Carbon dioxide $\rightarrow$ CO_2

Glucose $\rightarrow$ $C_6H_{12}O_6$

Methane $\rightarrow$ CH_4

Ammonia $\rightarrow$ NH_3

Sulphuric Acid $\rightarrow$ H_2SO_4

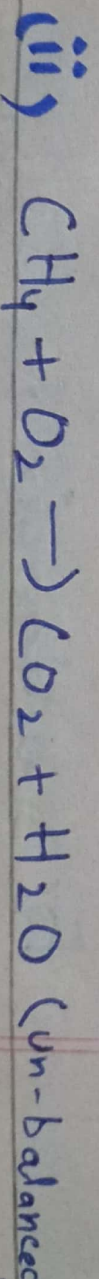
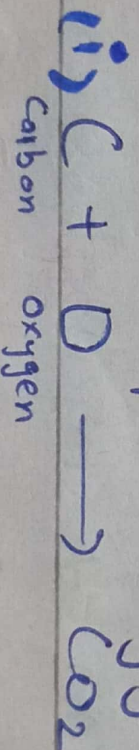
III Chemical Reaction from everyday life:

- (i) Burning
- (ii) Photosynthesis
- (iii) Respiration
- (iv) Milk $\xrightarrow{\text{Yeast}}$ Yogurt or Curd

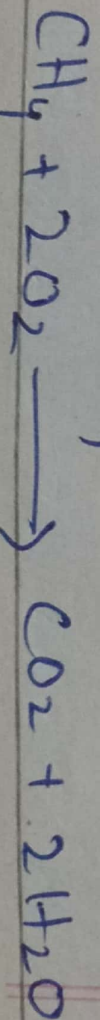
(i) Burning:-

$\therefore$ Coal burn in the

presence of oxygen)

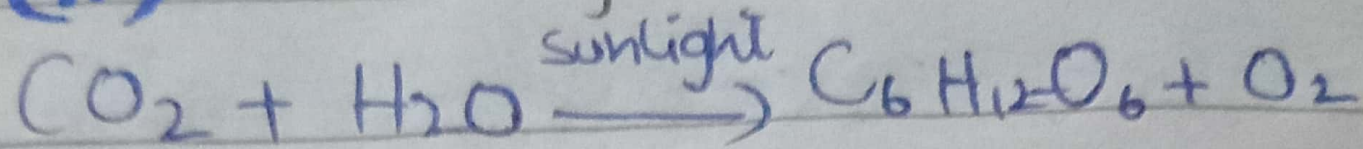


For balancing

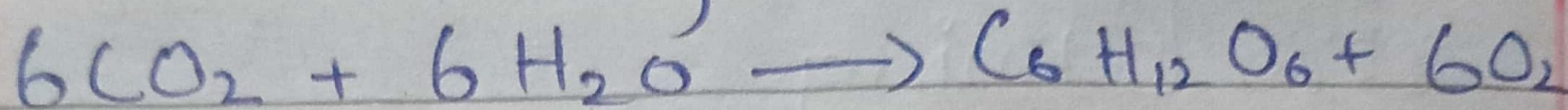


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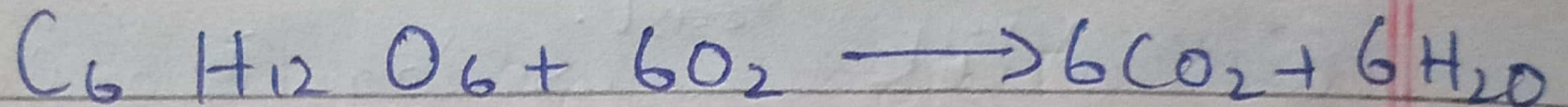
(ii) Photosynthesis:



For balancing



(iii) Respiration:



General Science

• Force & Motion:
✓ Movement & Motion

Motion:

If an object changes its position with respect to its surroundings then the body is said to be in Motion.

Example: (1) Moving Car
(2) Flying Bird. etc.

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Rest:

If an object does not change its position with respect to its surrounding then the body is said to be at rest.

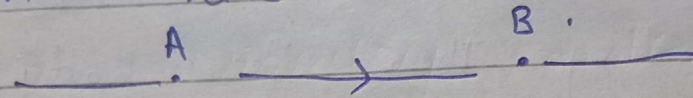
Example: (1) Tree
(2) Building etc.

Types of Motion:

(i) Translational Motion:

If a body moves from one point

to another point on a straight line. then the body is said to be in translational motion.



(2) Rotatory Motion/Spin Motion

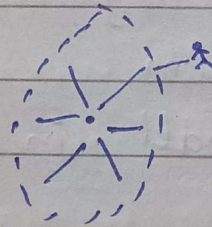
If a body rotates about a fixed axis the such a motion is called rotatory or spin motion.

Example: Top, Fan.

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(3) Orbital Motion / Circular

If an object moves on the circumference of a circle then this motion is called orbital motion.



(4) Oscillator / Vibratory Motion:

The motion that is executed around its mean position is called Oscillator or Vibratory motion.

Example:-

- (i) Swing
- (ii) Simple Pendulum
- (iii) Metal ball in a try.

(5) Random Motion:-

Zig zag motion is called Random Motion. Random Motion of gas is called Brownian motion.

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The disorder or irregular motion of an object.

For example:-

- (1) Insects
- (2) Birds
- (3) Atom and molecules

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General Science

Laws of Motions:-

∴ Newton's Law:-

1) First Law of Newton:-

If a body is at rest it will remain at rest until some external force is applied on it. and if a body is in motion it will keep on moving until some

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external force is
applied to stop
it.

For example: A book on the
Table. A moving car.

=> Rest:

If a body does not
change its position with
respect to its surroundings
it is called rest.

=> Motion:

Back

(2) Second Law of Motion

If external ^{force} is applied on an object which is at rest, acceleration ^{Rate of change of velocity} is produced in the body. This acceleration is directly proportional to the applied force and inversely proportional to the mass of the object.

$$a \propto F \quad \text{--- (1)}$$

$$a \propto \frac{1}{m} \quad \text{--- (2)}$$

Combine equations

$$a \propto \frac{F}{m} \quad \text{--- (3)}$$

$$a \propto F - 0$$

$$a \propto \frac{1}{m} \quad \text{--- (2)}$$

$$a \propto F$$

$$F = \frac{m}{k} \quad \text{--- (3)}$$

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$$a = k \frac{F}{m}$$

where $k=1$ Suppose $k=1$

$$a = \frac{F}{m} \quad ma = F \quad a = \frac{F}{m}$$

$$\boxed{F = ma} \quad F = ma \quad a = \frac{F}{m}$$

③ Third Law of Motion.

To every action,
there is an equal
reaction in
but opposite direction.

OR

Action and Reaction
are equal but in opposite
direction.

For e.g. bouncing the ball.
Rocket machine.

Force & Machines

(i) Force:-

The agency that moves a stationary object and stops a moving object is called Force.

(ii) Machine:-

An instrument that makes our work easy is called Machine.

There are Two Types of Machines

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- (i) Simple Machine
- (ii) Complex Machine

Simple Machines:

- (i) Lever
- (ii) Wheel & axle
- (iii) Wedge
- (iv) Screw
- (v) Pulley

Movable Pulley Fixed Pulley

In Movable Pulley In Fixed Pulley

Force & Force & Load

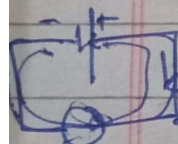
Load move in move in different
same directions. directions

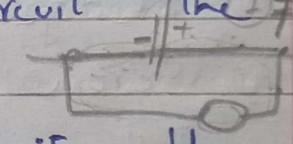
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Circuit:- any thing through which current passes or not

There are Two Types of Circuit:-

Open current $\rightarrow$ Rate of flow of charges through a conductor
Closed

In Closed circuit the $I \neq 0$
bulb is ON.  I is not zero

In Opened circuit the I is zero.
bulb is OFF.  I is zero.

$$\Rightarrow I = \frac{Q}{t}$$

where I is Electric Current, Q is electric charge, and t is time.

$\therefore$ Unit of electric Current is Ampere.

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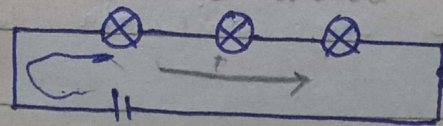
$$1 \text{ Ampere} = \frac{1 \text{ C}}{1 \text{ s}}$$

If 1 Coulomb charge flows through a circuit in 1s, then 1A current is present in the circuit.

Types of circuits:

- (i) Series circuit.
- (ii) Parallel circuit.

i) Series Circuit:



same voltage

same current

no path

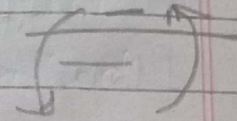
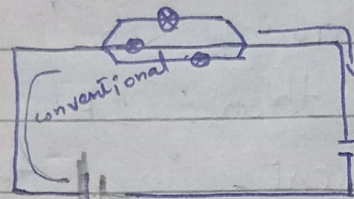
side to side

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different voltage

(iii) Parallel direction:

no same current



conventional
direction

It is always from
positive to negative terminal of
battery.

conventional $\rightarrow$ +ve to -ve
electrical $\rightarrow$ -ve to +ve

dimension

Resistance:

Opposition offered to
the flow of charges
is called Resistance.

Ohm law say unit is Ω

Ω

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Ohm's Law

$$V = IR$$

$$R = \frac{V}{I}$$

V → Voltage

I → Current

12V

10hm

Current measuring device

is Ammeter.

Current measuring device

Voltage / Potential → Voltmeter

Resistance → Ohmmeter

Current → Ammeter

1.1 Multimeter

A device which
measure Voltage, Resistance
& Current.

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Power:

Work done per unit Time is called Power

Formula: $P = \frac{W}{T}$

By multiplying Voltage and Current we have Power.

$$P = VI$$

$$V = IR$$

$V = \text{Voltage}$ $I = \text{Force}$

$I = \text{Electric Current}$ $M = \text{Mass}$

$R = \text{Resistance}$ $\alpha \text{ dire}$

$W = \text{Work}$ $d = \text{Acade}$

$T = \text{Time}$

$P = \text{Power}$

General Science

Energy:-

The capacity of a body to do work is called Energy.

Types of Mechanical Energy

(i) Kinetic Energy

It is due to the motion of the body.

(K.E)

$$K.E = \frac{1}{2} m v^2$$

mass velocity

$$K.E \propto m$$

$$K.E \propto v^2$$

$$K.E = \frac{1}{2} m v^2$$

(2) Potential Energy

It is due to the ^{position} potential of the body. (P.E)

$$P.E = m \underset{\text{gravity}}{g} \underset{\text{height}}{h}$$

Gravitational

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Magnetic field

Saturday

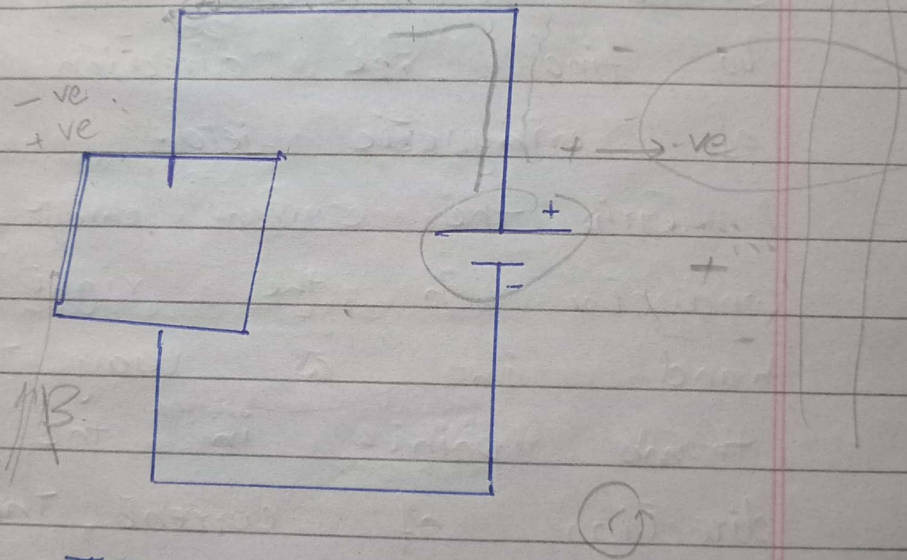
General Science.

Magnetic field

the space around a charge in which it exerts magnetic force

Electromagnetism

on other magnetic charges



There is magnetic field present around a current carrying conductor. Symbol of magnetic field = B